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```
import cv2
import numpy as np
from google.colab import files
from google.colab.patches import cv2_imshow
import matplotlib.pyplot as plt

uploaded = files.upload()

img_path = list(uploaded.keys())[0]

img = cv2.imread(img_path)

gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
```

Choose Files roi-dimor-B...nsplash.jpg

roi-dimor-BHA91Ia48kY-unsplash.jpg(image/jpeg) - 2956895 bytes, last modified:
9/15/2026 - 100% done

Saving roi-dimor-BHA91Ia48kY-unsplash.jpg to roi-dimor-BHA91Ia48kY-unsplas

```
import cv2
import numpy as np
from google.colab import files
from google.colab.patches import cv2_imshow
import matplotlib.pyplot as plt

plt.imshow(cv2.cvtColor(img, cv2.COLOR_BGR2RGB))
plt.title("Original Image")
plt.axis('off')
plt.show()
```

Original Image



```
import cv2
import numpy as np
from google.colab import files
from google.colab.patches import cv2_imshow
import matplotlib.pyplot as plt

gaussian = cv2.GaussianBlur(img, (51, 51), 0)

plt.figure(figsize=(12, 6))

plt.subplot(1, 2, 1)
plt.imshow(cv2.cvtColor(img, cv2.COLOR_BGR2RGB))
plt.title("Original")
plt.axis('off')

plt.subplot(1, 2, 2)
plt.imshow(cv2.cvtColor(gaussian, cv2.COLOR_BGR2RGB))
plt.title("Gaussian Blur")
plt.axis('off')

plt.show()
```

Original



Gaussian Blur



```
import cv2
import numpy as np
from google.colab import files
from google.colab.patches import cv2_imshow
import matplotlib.pyplot as plt

median = cv2.medianBlur(img, 31)

plt.figure(figsize=(12, 6))

plt.subplot(1, 2, 1)
plt.imshow(cv2.cvtColor(img, cv2.COLOR_BGR2RGB))
plt.title("Original")
plt.axis('off')

plt.subplot(1, 2, 2)
plt.imshow(cv2.cvtColor(median, cv2.COLOR_BGR2RGB))
plt.title("Median Filter")
plt.axis('off')

plt.show()
```

Original



Median Filter



```
# Average Filter
```

```
average = cv2.blur(img, (25, 25))
```

```
plt.figure(figsize=(12, 6))
```

```
plt.subplot(1, 2, 1)
```

```
plt.imshow(cv2.cvtColor(img, cv2.COLOR_BGR2RGB))
```

```
plt.title("Original")
```

```
plt.axis('off')
```

```
plt.subplot(1, 2, 2)
```

```
plt.imshow(cv2.cvtColor(average, cv2.COLOR_BGR2RGB))
```

```
plt.title("Average Filter")
```

```
plt.axis('off')
```

```
plt.show()
```

Original



Average Filter



```
import cv2
import numpy as np
from google.colab import files
from google.colab.patches import cv2_imshow
import matplotlib.pyplot as plt

laplacian = cv2.Laplacian(gray, cv2.CV_64F)
laplacian = cv2.convertScaleAbs(laplacian)

plt.figure(figsize=(12, 6))

plt.subplot(1, 2, 1)
plt.imshow(cv2.cvtColor(img, cv2.COLOR_BGR2RGB))
plt.title("Original")
plt.axis('off')

plt.subplot(1, 2, 2)
plt.imshow(laplacian, cmap='gray')
plt.title("Laplacian")
plt.axis('off')

plt.show()
```

Original



Laplacian



```
import cv2
import numpy as np
import matplotlib.pyplot as plt

sobel_x = cv2.Sobel(gray, cv2.CV_64F, 1, 0, ksize=3)
sobel_y = cv2.Sobel(gray, cv2.CV_64F, 0, 1, ksize=3)

sobel_x = cv2.convertScaleAbs(sobel_x)
sobel_y = cv2.convertScaleAbs(sobel_y)

sobel_combined = cv2.bitwise_or(sobel_x, sobel_y)

plt.figure(figsize=(16, 5))

plt.subplot(1, 3, 1)
plt.imshow(sobel_x, cmap='gray')
plt.title("Sobel X")
plt.axis('off')

plt.subplot(1, 3, 2)
plt.imshow(sobel_y, cmap='gray')
plt.title("Sobel Y")
plt.axis('off')

plt.subplot(1, 3, 3)
plt.imshow(sobel_combined, cmap='gray')
plt.title("Sobel Combined")
plt.axis('off')

plt.show()
```

Sobel X



Sobel Y



Sobel Combined



```
print("Observations:")

print("- Gaussian Blur: Noise kam karta hai, image ko smooth banata hai,
print("- Median Filter: Salt-and-pepper noise ko bahut achhe se hatata ha
print("- Average Filter: Simple smoothing karta hai, lekin edges Gaussian
print("- Laplacian Filter: Image ke edges aur fine details ko highlight k
print("- Sobel X/Y: Horizontal aur vertical gradients detect karta hai, e
print("- Sobel Combined: Dono directions ke edges ek saath dikhata hai.")
```

Observations:

- Gaussian Blur: Noise kam karta hai, image ko smooth banata hai, edges th
- Median Filter: Salt-and-pepper noise ko bahut achhe se hatata hai, edges
- Average Filter: Simple smoothing karta hai, lekin edges Gaussian se zyad
- Laplacian Filter: Image ke edges aur fine details ko highlight karta hai
- Sobel X/Y: Horizontal aur vertical gradients detect karta hai, edges ki
- Sobel Combined: Dono directions ke edges ek saath dikhata hai.

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

gaussian = cv2.GaussianBlur(img, (25, 25), 0)
median = cv2.medianBlur(img, 15)
average = cv2.blur(img, (25, 25))

laplacian = cv2.Laplacian(gray, cv2.CV_64F)
laplacian = cv2.convertScaleAbs(laplacian)

sobel_x = cv2.Sobel(gray, cv2.CV_64F, 1, 0, ksize=3)
sobel_y = cv2.Sobel(gray, cv2.CV_64F, 0, 1, ksize=3)

sobel_x = cv2.convertScaleAbs(sobel_x)
sobel_y = cv2.convertScaleAbs(sobel_y)

sobel_combined = cv2.bitwise_or(sobel_x, sobel_y)
```

```
titles = [  
    'Original',  
    'Gaussian Blur',  
    'Median Filter'.  
]
```